VAISHAK V

Mechatronics Engineer

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CAREER OBJECTIVE

Robotics-focused engineering student with hands-on experience in ROS2, Python, C++, Ubuntu/Linux and robotic simulation. Interested in autonomous systems, robot perception, navigation, control and sensor integration. Experienced in developing and testing robotics projects using simulation and software tools, with a strong interest in applying robotics technologies to real-world systems.

TECHNICAL SKILLS

Robotics & Simulation: ROS2,ROS2 (Concepts), Gazebo, RViz, Robot Kinematics (transformations, DH parameters basics), SLAM, Nav2

Design Tools: SolidWorks, Fusion 360

Programming & OS: C, C++ , Python, Linux/CLI

Hardware & Systems: Arduino, Raspberry, Sensors, Motors

PROJECTS

Mecanum-Wheeled Obstacle Avoidance Robot

Technologies: Arduino, C/C++, Ultrasonic Sensors, Motor Control

- Designed and developed a mecanum-wheeled robot for autonomous obstacle avoidance.
- Integrated ultrasonic sensors and motor-control logic for real-time obstacle detection and movement.
- **Winner – BIT Hackathon Season 2, Robotics & Automation.**

Automated Waste Sorting Conveyor System

Technologies: Automation, Sensors, Arduino, Motor Control

- Designed an automated conveyor system for detecting and sorting waste materials.
- Integrated sensors, actuators, and control logic for automated sorting operations.

Autonomous Mobile Robotics (AMR) – WorldSkills

Technologies: Raspberry Pi, Python, Sensors, Computer Vision, Gripper

- Developed an AMR using Raspberry Pi with sensor-based line following and motor control.
- Implemented color detection and autonomous object pick-and-place using a gripper.
- **Advanced to Stage 2 – WorldSkills, Autonomous Mobile Robotics (AMR).**

Smart Pressure-Mapped Air-Bladder Support System – Aakruti

Technologies: FSR Sensors, Pressure Mapping, Embedded Systems, Pneumatic Actuation

- Designed a virtual concept model using **FSR-based 8-zone pressure mapping** for localized pressure monitoring.
- Proposed an **automatic air-bladder support mechanism** based on detected pressure distribution.
- **National Semifinalist – Aakruti Design Challenge.**

ACHIEVEMENTS

- **National Semi-finalist —** Aakruti Global Design Challenge 2025
- **Regional Finalist —** TN Skills (Autonomous Mobile Robotics)
- **ROBOCON 2025 & 2026 — Design Team**
  - **2025:** Collaboratively designed and optimized basketball robot mechanisms using **Fusion 360**, contributing to autonomous operation and robot performance.
  - **2026:** Designed the **R2 robot** for the Kung Fu-themed challenge, focusing on mechanisms for **terrain traversal and competition tasks** in Solidworks.

EDUCATION

- **B.E. Mechatronics Engineering,** Bannari Amman Institute of Technology (2024–Present) — CGPA: 8.15